

Traceability Matrix

Terminology used in "Tested by":

QT console = QT creator IDE console. Console on the GUI is mostly for feedback on button clicks.

10 second mark = My system has a single 10 second timer that moves all elevators (if there are requests) to simulate concurrency. The timer starts as soon as the program runs and repeats forever.

Requirements	Related Use Case	Implemented by	Tested by
A building is serviced by M elevators	N/A	ElevatorControlSystem, Elevator	See ElevatorControlSystem's constructor for variability in elevators.
The elevator (car) can move from floors 1 to N	Using an elevator	ElevatorControlSystem, Elevator, Floor	See ElevatorControlSystem's constructor for variability in floors. Run the program and using the GUI: select a passenger, request an elevator going up, enter the elevator after you see the 10 second mark, request a floor, see the elevator cart move after another 10 second mark.
Passengers can request the elevator going in a direction with a panel marked "up" and "down" on each floor	Using an elevator	MainWindow, Controller, ElevatorControlSystem, Floor, UpAndDownPanel, Passenger	Run the program and using the GUI: select a passenger and click on "request up" or "request down"
Passengers can exit and enter elevators	Using an elevator	MainWindow, Controller, Passenger, Floor, Elevator, Door,	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and the passenger can enter and exit with "enter" and "exit" buttons.
Buttons illuminate when pressed and	Using an elevator	MainWindow, Controller, Elevator, ElevatorPanel	Run the program and using the GUI: select a passenger, click on "request up", after

remains so until the elevator arrives at the floor			10 seconds, an elevator will open and enter the elevator using the "enter" button. Then click on one of the floor buttons and see the buttons illuminate blue when clicked. When the elevator arrives at the floor, the blue will disappear and the button will be grey again.
The elevator door rings when opening and closing	Using an elevator	ElevatorControlSystem, Elevator	Run the program and using the GUI: select a passenger and click on "request up." After 10 seconds, see the QT console simulate a door ring with a console message.
The elevator door remains open for 10 seconds when it arrives at a floor	Using an elevator	MainWindow, Controller, ElevatorControlSystem, Elevator	Run the program and using the GUI: select a passenger and click on "request up." After the first 10 second mark, the 10 second timer will be reset and an elevator door will be open. The repeat 10 second timer simulates a door being open for 10 seconds.
The elevator has a display that shows the current floor of the elevator	Using an elevator	MainWindow, Controller, ElevatorControlSystem, Elevator	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and enter the elevator using the "enter" button. The elevator panel in the GUI will display "Floor: 1" instead of "Floor: -1" to display the current floor of the elevator that the selected passenger is in.
The elevator control panel can manually close and open doors with respective buttons	Using an elevator	MainWindow, Controller, Elevator, ElevatorPanel, Door	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and enter the elevator using the "enter" button. Usually, the elevator remains open until the next 10

			second mark and the passenger can enter and exit freely using the "exit" and "enter" buttons. If the passenger is inside the elevator, they can click the close door button to close the elevator door and the passenger cannot exit the elevator. Similarly, they can open the door using the open door button to exit the elevator.
The close door button can override the default 10 second timing	Using an elevator	MainWindow, Controller, Elevator, ElevatorPanel, Door	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and enter the elevator using the "enter" button. Usually, the elevator remains open until the next 10 second mark and the passenger can enter and exit freely using the "exit" and "enter" buttons. If the passenger is inside the elevator, they can click the close door button to close the elevator door and they would not be able to leave unless they click the open door button.
Each elevator has a sensor that notifies it when it arrives at a floor	Using an elevator	ElevatorControlSystem, Elevator, Floor	The sensor is implemented in the system with the elevator's senseFloor() function which simply updates the floor object of the elevator (floor object is an attribute of the elevator that connects the elevator and the current floor it's on). Run the program and using the GUI: select a passenger, click on "request up", after the next 10 second mark, an elevator will open and click "enter." Click on any of the elevator floor buttons and

			wait for the next 10 second mark for the elevator to move to the desired floor. Click "exit" to exit the elevator and see that the selected passenger is on the new floor. The passenger being able to move between the elevator and the floor demonstrates that the elevator has the correct pointer to the floor object.
The elevator control system ensures that the group of elevators services all requests expeditiously	N/A	ElevatorControlSystem, Elevator, Floor, ElevatorPanel	Examine the code of the elevator system and/or test using the GUI. In the GUI, select different passengers and use the elevators to go to different floors. Different scenarios can be performed to see that the ElevatorControlSystem's algorithm services requests efficiently as possible.
The elevator has an audio system to announce messages/warnings	The light sensor is interrupted, The "fire" alarm, The "overload" alarm, The "power out" alarm	ElevatorControlSystem, Elevator	Run the program and using the GUI: click on the "fire" or "power out" buttons in the emergency testing section. In the QT console, you can see messages being played for all elevators.
There is a help button inside the elevator linked to building safety service and 911. When pressed, a passenger is connected to building safety services. If there is no response within 5 seconds, 911 is	The "help" alarm	ElevatorControlSystem, ElevatorPanel	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and enter the elevator using the "enter" button. Click the "help" button and watch the QT console. The QT console will first print out a message indicating a connection to building safety services is being attempted. After 5 seconds, QT console will print that emergency

called			services is being called.
Door obstacles prevent closing of the doors of an elevator and the elevator does not move because of this	The light sensor is interrupted	MainWindow, Controller, ElevatorControlSystem, Elevator, Door	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and enter the elevator using the "enter" button. In the "elevator testing" section, select the elevator that the passenger is in using the dropdown box. Then click the "block door" button. After 10 seconds, QT console will print a message indicating that the elevator cannot move due to an obstacle that prevents the closing of the elevator door.
The fire alarm signal causes all the elevators to go to a safe floor , warning messages are made, and passengers can exit on the safe floor	The "fire" alarm	MainWindow, Controller, ElevatorControlSystem, Elevator, Passenger	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and enter the elevator using the "enter" button. Next, move to any floor that's not the current one using the elevator panel. Once on a different floor, click either the fire button in the elevator panel or the fire button in the emergency testing section. This will print messages relating to the fire signal being activated,in the QT console. After the next 10 second mark, the selected passenger's elevator should be at the first floor and any other elevators should be too. A message for each elevator should be printed to the QT console, asking the passenger to exit the elevator.
If an elevator exceeds its weight limit,	The "overload" alarm	ElevatorControlSystem, Elevator, Door, Passenger	Run the program and using the GUI: select passenger 1, click on "request up", after

then it does not move and asks passengers to reduce the load before attempting to move again			10 seconds, an elevator will open and enter the elevator using the "enter" button. Before the next 10 second mark, select passenger 2 and enter the elevator as well. The weights of both passengers should be greater than 500, seen in the elevator testing section when you select Elevator 1 in the dropdown box. In the next 10 second mark, the ecs should attempt to move all elevators but should fail for elevator 1. A QT console message should be printed to ask passengers to reduce the load.
If the power out signal has been received, all elevators will move to a safe floor and ask passengers to exit on that floor	The "power out" alarm	MainWindow, Controller, ElevatorControlSystem, Elevator, Passenger	Run the program and using the GUI: select a passenger, click on "request up", after 10 seconds, an elevator will open and enter the elevator using the "enter" button. Next, move to any floor that's not the current one using the elevator panel. Once on a different floor, click on the "power out" button in the emergency testing section. This will print messages relating to the power out signal being activated, in the QT console. After the next 10 second mark, the selected passenger's elevator should be at the first floor and any other elevators should be too. A message for each elevator should be printed to the QT console, asking the passenger to exit the elevator.